

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – ECOLOGY (1.2): FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? You have watched the Kenyan river video (2:20 to 4:50), investigated habitats, adaptations, populations, sampling methods, limiting factors, nutrient cycles and pollution. Now write your own Final Explanation. Answer each section in the space under the prompt, using evidence from your lessons, your presentations, your reading summary, your cycle drawings and your pollution summary. Use correct ecological terms and Kenyan examples. Write in full sentences. Where asked, you may add a labelled sketch or a simple calculation.

Section 1: The River as an Ecosystem

At the start, people thought the fish were old or had no food. (a) Using what you learned in Lesson 1, explain why these ideas are not enough to explain what is happening in the river. (b) Use at least four ecological terms (for example habitat, population, community, niche, biotic factor, abiotic factor) to describe the river. (c) Explain what the phrase 'the river is still full of life' tells you about how organisms interact with their environment.

(a) If fish were only old or short of food, we would not expect algae and aquatic weeds to boom or the water to smell. These ideas explain only part of the evidence, so something else in the environment must be changing. (b) The river is an ecosystem: a community of living organisms interacting with each other and with non-living factors. Its habitat is the water and banks where organisms live. All the tilapia (fish of one species) in the river form a population; the fish, algae, weeds and bacteria together form the community. Each species has a niche, its role in the river. Biotic factors include fish, algae, weeds and decomposers; abiotic factors include water, light, temperature and dissolved oxygen. (c) The river still has plenty of living things, but not the same ones as before. Organisms depend on the environment and on each other, so a change in one part, including change caused by human activity, can help some organisms (algae, weeds) while harming others (fish).

Section 2: Adaptations and Habitats

(a) Choose two organisms from different habitat groups that you studied (for example Euphorbia, maize, water hyacinth, mangroves, camel, tilapia). For each, name its

(a) Euphorbia is a xerophyte: it has a thick waxy cuticle and spines (reduced leaves), which reduce water loss in dry places. Tilapia is an aquatic animal: it has gills that take dissolved oxygen from the water, so it can breathe underwater. (Other correct pairs are accepted, e.g. camel with water-conserving kidneys, mangroves with salt glands, water hyacinth with air spaces for floating.) (b) Water hyacinth and other aquatic weeds are hydrophytes, plants adapted to living in water, with air spaces that help them float. Algae are protists, not hydrophyte plants. (c) Warm water and extra nutrients favour the growth of algae and aquatic weeds. Fish use gills to get dissolved oxygen, so they need enough oxygen in the water. If the environment changes so that oxygen is low,

habitat group and describe one adaptation and how it helps the organism survive. (b) Water hyacinth and algae both grow thickly in the river. Explain which of them is a hydrophyte and which is not. (c) Explain why conditions in the river can favour weeds and algae but harm fish. Link your answer to the adaptations of fish.	fish are harmed even though algae and weeds do well. Each organism is adapted to certain conditions, so a change in the environment helps some and harms others.
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Section 3: Measuring Populations	
(a) Define population and population density. (b) In the school compound your group counted every individual in a 10 m x 10 m area. Explain when direct counting is suitable and why it would be difficult in the river. (c) Describe how quadrat sampling (1 m x 1 m quadrats placed randomly) and a belt transect helped you find out whether plants are spread evenly. State what you found and what caused the differences. (d) In a capture-recapture simulation, 20 counters were marked and returned to the container. A second sample of 20 contained 5 marked counters. Calculate the estimated population and compare it with the true value of 100. State two assumptions of the method and say how the method could help the community know if fish numbers are falling.	(a) A population is all the individuals of one species living in an area. Population density is the number of individuals per unit area (number / area). (b) Direct counting is suitable when the population is small and the individuals are easy to see, identify and count, such as grass tufts or weeds in a marked area. Fish are hidden under water, move around and cannot be counted one by one, so direct counting is not possible in the river. (c) Quadrats placed randomly let us count plants in small areas, find the mean number per quadrat, calculate density and estimate the total population without counting every plant. A belt transect, with two parallel lines and quadrats at regular intervals, let us record plant numbers across the open field, shade, pathway and garden. Plants were not spread evenly: numbers differed between sunny, shaded, damp, trampled and compacted spots, because light, water, soil and trampling affect how many plants can live in a place. (d) Estimate = $(20 \times 20) / 5 = 80$. The true value is 100, so the estimate is close but a little low; more trials and pooled class data give a more reliable estimate. Assumptions: the marked and unmarked individuals are thoroughly mixed; there are no births, deaths or migration between samples; marks are not lost. Repeating the method over time would show whether the estimated number of fish is falling, although it does not tell us why.

Section 4: Why Are the Fish Disappearing?

Use your reading summary on limiting factors, your nitrogen and carbon cycle drawings, and your pollution summary. (a) Name two biotic and two abiotic factors that can limit a population, and explain what carrying capacity means. (b) Describe how nitrogen and nutrients from a place such as a maize shamba, a dairy farm near Kericho or sewage can reach the river. (c) Write a step-by-step explanation, in order, of how this leads to the booming of algae and weeds, the bad smell and the loss of fish. (d) Explain how garbage and effluent, as in the Nairobi and Athi Rivers, add to the problem.	(a) Biotic factors: predation, competition, disease, food supply. Abiotic factors: temperature, dissolved oxygen, light, nutrients, water level. Carrying capacity is the largest population an environment can support; populations stop growing when limiting factors balance growth, and factors interact with each other. (b) Fertiliser, manure and sewage are washed by rain as runoff into streams and then into the river that flows towards Lake Victoria. Nitrogen and other nutrients move from soil and waste into water instead of being taken up and recycled on land. (c) 1. Extra nutrients (nitrates and phosphates) enter the river from farms and settlements. 2. Algae and aquatic weeds grow very fast (algal bloom, eutrophication). 3. Thick growth reduces the light reaching the water below. 4. When algae and weeds die, bacteria and other decomposers break them down and use up dissolved oxygen. 5. The decomposition produces foul-smelling gases, so the water smells. 6. With little oxygen, fish with gills suffocate or leave, so fewer fish are caught. Overfishing, predators and competition may add to the decline, but the nutrient pollution explains the algae, weeds and smell. (d) Effluent and sewage from towns add nutrients and harmful substances, while garbage such as plastic and polythene and oil and detergent from car washes add harmful substances; garbage can also block drains and the flow of water. This worsens low oxygen and kills or drives away organisms. Burning waste and fossil fuels also adds greenhouse gases that contribute to global warming and climate change.
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Section 5: Restoring Our River

The driving question asks what the community can do to restore the ecosystem. (a) Propose three actions the community could take, and for each explain which part of the problem it addresses. (b) Describe one way you could check, using a method from this unit, whether your actions are working. (c) Compare your first model of the river in Lesson 1 with what you now know. State two things you would add or change.	(a) 1. Manage waste properly (collect solid waste, unblock drains, stop dumping plastic and polythene): this reduces garbage and pollutants reaching the river. 2. Treat effluent and control fertiliser and manure runoff from farms and settlements: this reduces nitrates and phosphates, so fewer algal blooms and less oxygen is used up by decomposers. 3. Plant vegetation on the riverbank: plants take up nutrients and reduce runoff. Reducing pollution from burning rubbish and vehicles also helps. (b) Use capture-recapture to estimate the fish population over time: $\text{estimated population} = (\text{number marked} \times \text{number in second sample}) / \text{number recaptured}$. If estimates rise over several months, the fish are recovering. Quadrat or belt transect counts on the bank vegetation could also show how plant populations change, and we can notice whether the smell and thick algae decrease. (c) My first model showed only the river, fish and plants. I would add: the nutrient pathway from farms, sewage and garbage into the river; algae and weeds growing then dying; bacteria using up dissolved oxygen and producing smell; fish dying or leaving. I would also add human activity as a factor that links the land to the river.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Use of evidence from the lessons	Uses specific evidence from the river video, the school field investigations, simulations, readings and pollution research to support every claim.	Uses some evidence from the lessons, but some claims are general or unsupported.	Gives few or no pieces of evidence, or evidence is not linked to the river.
Scientific accuracy and ecological terms	Uses terms such as ecosystem, habitat, population, density, carrying capacity, eutrophication and dissolved oxygen correctly, with accurate adaptations and calculations.	Uses most terms correctly with minor errors in adaptations or calculations.	Uses few terms or makes major errors, such as confusing algae with hydrophytes or population with community.
Causal explanation of the phenomenon	Gives a complete ordered chain from nutrient runoff to algal bloom, decomposition, low oxygen, smell and fish loss, and mentions other limiting factors.	Gives a mostly correct chain but misses a step or does not link the smell or oxygen.	Lists causes without explaining how they connect, or relies on the original ideas (old fish, no food).
Sampling methods and data	Explains direct counting, quadrats, belt transects and capture-recapture correctly, calculates the estimate accurately and states assumptions.	Describes methods and calculates correctly, but explanations of when to use them or the assumptions are incomplete.	Confuses methods or cannot calculate the estimate.
Community solutions and model revision	Proposes realistic actions that each target a cause, suggests a way to monitor results, and clearly shows how the model has improved.	Proposes reasonable actions and some model changes, but links to causes or monitoring are weak.	Gives vague or unrelated actions and little model revision.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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